

CPU Benchmarks

Over 1,000,000 CPUs Benchmarked



AMD Ryzen 9 9950X3D

Benchmarks for the AMD Ryzen 9 9950X3D can be found below. Release dates, price and performance comparisons are also listed when available. This is made using thousands of [PerformanceTest](#) benchmark results and is updated daily.

- The first graph shows the relative performance of the CPU compared to the 10 other common (single) CPUs in terms of PassMark CPU Mark.
- The 2nd graph shows the value for money, in terms of the CPUMark per dollar.
- The pricing history data shows the price for a single Processor. For multiple Processors, multiply the price shown by the number of CPUs.

Is the AMD Ryzen 9 9950X3D Good for Gaming?

This is relatively new CPU that contains 16 Cores and 32 Threads. The AMD Ryzen 9 9950X3D also has great single threaded performance that will serve well in games. Paired with a good [videocard](#), *is this CPU Good for Gaming?* Yes, this is a high end chip that would be great for gaming. [\[More...\]](#)



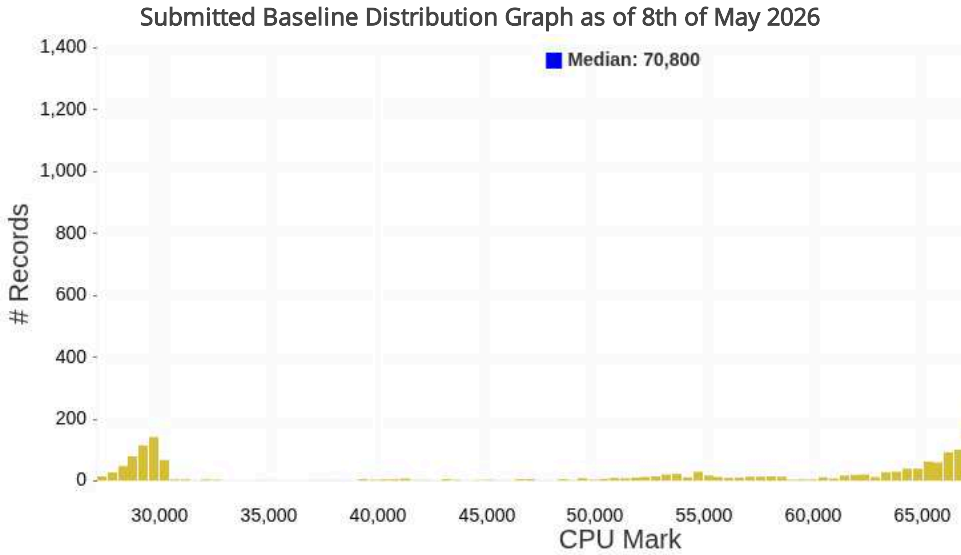
CPUS	AMD Ryzen 9 9950X3D	Average CPU Mark 
 Multithread Page: 1, 2, 3, 4 Single Thread Page: 1, 2, 3, 4	Description: AMD Radeon Graphics	Multithread Rating 70190
 Best Value (On Market) Best Value XY Scatter Best Value (All time)	Class: Desktop Socket: AM5 Clockspeed: 4.3 GHz Turbo Speed: 5.7 GHz Cores: 16 Threads: 32 Cache per CPU Package: L1 Instruction Cache: 16 x 32 KB L1 Data Cache: 16 x 48 KB L2 Cache: 16 x 1024 KB L3 Cache: 128 MB	Single Thread Rating 4743 Samples: 9761* * Margin for error: Low
 New Desktop New Laptop New Server New Mobile	Other names: AMD Ryzen 9 9950X3D 16-Core Processor CPU First Seen on Charts: Q1 2025 CPUMark/\$Price: 111.59	+ COMPARE
 Systems with Multiple CPUs Overclocked Power Performance CPU Mark by Socket Type Cross-Platform CPU Performance Top Gaming CPUs		

CPU Test Suite Average Results for AMD Ryzen 9 9950X3D

Integer Math	243,078 MOps/Sec
Floating Point Math	159,707 MOps/Sec
Find Prime Numbers	609 Million Primes/Sec
Random String Sorting	95,362 Thousand Strings/Sec
Data Encryption	44,475 MBytes/Sec
Data Compression	907,452 KBytes/Sec
Physics	5,665 Frames/Sec
Extended Instructions	72,962 Million Matrices/Sec
Single Thread	4,743 MOps/Sec

From submitted results to PerformanceTest V10 as of 12th of May 2026.

CPU Mark Distribution for AMD Ryzen 9 9950X3D



From submitted results since PerformanceTest V10 as of 8th of May 2026.
For distribution graph only: Results are [trimmed](#) to exclude outliers by disregarding the bottom 1% and top 1% of submissions.



Search for AMD Ryzen 9 9950X3D



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Machines with this CPU (or similar)



[Skytech Gaming Legacy 4 Gaming PC. AMD Ryzen 9 9950X3D 4.3GHz, NVIDIA RTX 5090 32GB VRAM, X870 Board, 2TB Gen5 NVMe SSD, 64GB DDR5 RAM 6000, 1200W Gold ATX 3 PSU, 420 ARGB AIO, WI-FI 7, Windows 11](#)

\$5999.99
(www.amazon.com)



[STORMCRAFT Phantom RTX 5080. AMD Ryzen 7 9800X3D, 32GB DDR5 RAM 6000MHz, 2TB NVMe Gen4 SSD, B850 Chipset 850w PSU 360mm AIO, Win 11 Home, RGB Keyboard Mouse, WiFi BT HDMI AI Prebuilt Gaming Desktop PC](#)

\$2999.99
(www.amazon.com)



[Thermaltake LCGS View 9580S-380XL Gaming Desktop \(AMD Ryzen™ 9 9950X3D, ToughRam 32GB DDR5 6000MT/s RGB Memory, NVIDIA® GeForce RTX™ 5080, 2TB NVMe M.2, WiFi, Windows 11\) VXL-S-X870-580-LCS](#)

\$3899.99
(www.amazon.com)



[Mantis V2 - RTX 5070, AMD Ryzen 9 9950X3D Processor, 32GB DDR5 RAM, 1TB NVMe Gen4 SSD, Gaming Desktop PC, WiFi 7 + BT, RGB Gamer Keyboard+Mouse, 3X DP, HDMI, Windows 11 Pro - Best Prebuilt Computer](#)

\$2599.99
(www.amazon.com)



[Skytech Gaming Prism 4 Desktop PC. Ryzen 9 9950X3D 4.3 GHz \(5.7GHz\), NVIDIA RTX 5080 16GB, X870 Board, 4TB Gen4 NVMe SSD, 64GB DDR5 RAM 6000 RGB, 1000W Gold ATX 3 PSU, 360mm ARGB AIO, Wi-Fi, Win 11](#)

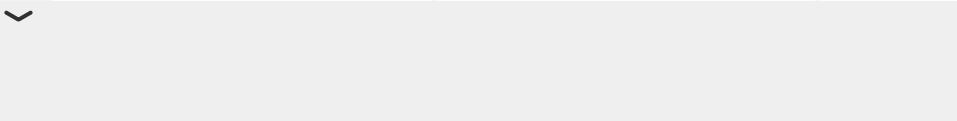
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CPU Mark Relative to Top 10 Common Desktop CPUs

As of 12th of May 2026 - Higher results represent better performance

Processor	Average CPU Mark	
AMD Ryzen 9 9950X3D		70,190
Intel Core i9-14900K		58,357
Intel Core i9-13900K		58,196
AMD Ryzen 9 5900X		38,920
Apple M4 10 Core		23,656
AMD Ryzen 5 5600X		21,838
Intel Core i5-12400F		19,588
Intel Core i9-9900K @ 3.60GHz		18,064
AMD Ryzen 5 3600		17,668
Apple M2 8 Core 3500 MHz		15,644



Processor	CPU Mark / \$Price	
AMD Ryzen 5 3600		170.53
AMD Ryzen 5 5600X		139.37
AMD Ryzen 9 5900X		126.50
Intel Core i9-14900K		124.43
Intel Core i5-12400F		122.43
AMD Ryzen 9 9950X3D		111.59
Intel Core i9-13900K		107.77
Intel Core i9-9900K @ 3.60GHz		69.70
Apple M1 8 Core 3200 MHz		NA
Apple M2 8 Core 3500 MHz		NA
Apple M4 10 Core		NA

Single Thread Rating

As of 12th of May 2026 - Higher results represent better performance

Processor	Average Thread Rating	
AMD Ryzen 9 9950X3D		4,743
Intel Core i9-14900K		4,690
Intel Core i9-13900K		4,598
Apple M4 10 Core		4,509
Apple M2 8 Core 3500 MHz		3,883
Apple M1 8 Core 3200 MHz		3,674
Intel Core i5-12400F		3,489
AMD Ryzen 9 5900X		3,466
AMD Ryzen 5 5600X		3,366
Intel Core i9-9900K @ 3.60GHz		2,912
AMD Ryzen 5 3600		2,559



Baseline	CPU Mark
BL3461230 - May 10 2026	71528
BL3461207 - May 10 2026	71283
BL3461138 - May 10 2026	73212
BL3461131 - May 10 2026	70064
BL3461059 - May 10 2026	71514

Additional baselines can be obtained using Windows version of [PerformanceTest's Manage Baselines](#) feature.

Popular comparisons for AMD Ryzen 9 9950X3D

As of 12th of May 2026 - Higher results represent better performance

Processor	Average CPU Mark
AMD Ryzen 9 9950X3D	70,190
AMD Ryzen 9 9950X vs AMD Ryzen 9 9950X3D	65,745 (-6.3%)
AMD Ryzen 9 9900X3D vs AMD Ryzen 9 9950X3D	56,223 (-19.9%)
Intel Core Ultra 9 285K vs AMD Ryzen 9 9950X3D	67,287 (-4.1%)
AMD Ryzen 9 7950X3D vs AMD Ryzen 9 9950X3D	62,318 (-11.2%)
AMD Ryzen 9 7950X vs AMD Ryzen 9 9950X3D	62,192 (-11.4%)
Intel Core i9-13900KS vs AMD Ryzen 9 9950X3D	60,509 (-13.8%)
Intel Core i9-13900K vs AMD Ryzen 9 9950X3D	58,196 (-17.1%)
Intel Core Ultra 9 285 vs AMD Ryzen 9 9950X3D	57,542 (-18.0%)
AMD Ryzen 9 9900X vs AMD Ryzen 9 9950X3D	54,416 (-22.5%)
AMD Ryzen 9 9955HX3D vs AMD Ryzen 9 9950X3D	62,642 (-10.8%)
Intel Core i9-14900KS vs AMD Ryzen 9 9950X3D	60,189 (-14.2%)
Intel Core Ultra 7 265K vs AMD Ryzen 9 9950X3D	58,693 (-16.4%)

Is the AMD Ryzen 9 9950X3D Good for Gaming?



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AMD Ryzen 7 9850X3D		12,507
AMD Ryzen 9 9900X3D		12,421
AMD Ryzen 7 9800X3D		12,121
AMD Ryzen 9 9950X3D		11,264
Intel Core Ultra 5 250K Plus		11,180
Intel Core Ultra 7 270K Plus		10,941
Intel Core Ultra 5 250KF Plus		10,849
AMD Ryzen 5 7600X3D		10,519
AMD Ryzen 9 7900X3D		10,384
Intel Core Ultra 9 285K		10,365
AMD Ryzen 5 7500X3D		10,133
AMD Ryzen 7 7800X3D		9,849

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